



① c_i : contribution at time 0

② $a_i = (1 + r_i) c_i$ accumulated wealth at time 1.
could be random

Let $I_i = \begin{cases} 1 & \text{if individual } i \text{ experiences the predefined event before time 1} \\ 0 & \text{otherwise} \end{cases}$

!! $\rightarrow$ **Survival** is the predefined event used below
 (could be being diagnosed with a critical illness, medical treatments, compensation for lost income, etc).

(#) Let $X_i = a_i(1 - I_i)$ and $S_n = \sum_{i=1}^n X_i$

(#) Consider $H_n = \left\{ (Y_1, \dots, Y_n) \in Y^n : \sum_{i=1}^n Y_i = S_n \right\}$
set of sharing-rules

and $H_n^{A-F} = \left\{ (Y_1, \dots, Y_n) \in H_n : E[Y_i] = E[X_i] \forall i \right\}$
set of actuarially fair sharing-rules

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Δ convex combination: Let $(Y_1^{(1)}, \dots, Y_n^{(1)}), \dots, (Y_1^{(k)}, \dots, Y_n^{(k)})$ denote k sharing-rules from H_n^{A-F} and $Y_i^1 = \sum_{k=1}^k \lambda_k Y_i^{(k)}$ with $\sum_{k=1}^k \lambda_k = 1$. Then $(Y_1^1, \dots, Y_n^1) \in H_n^{A-F}$

Proof:

$$\begin{aligned} \textcircled{\bullet} \sum_{i=1}^n Y_i^1 &= \sum_{i=1}^n \sum_{k=1}^k \lambda_k Y_i^{(k)} \\ &= \sum_{k=1}^k \lambda_k \left(\sum_{i=1}^n Y_i^{(k)} \right) \rightarrow = S_n \quad \forall k \end{aligned}$$

$$= S_n$$

$$\left(\sum_{k=1}^n \lambda_k = 1 \right)$$

$$\Rightarrow (Y_1^1, \dots, Y_n^1) \in H_n$$

$$\textcircled{\bullet} E[Y_i^1] = \sum_{k=1}^k \lambda_k E[Y_i^{(k)}]$$

$$= E[X_i]$$

$$\textcircled{\bullet} \text{ Let } W_i^1 = \begin{cases} a_i + Y_i^1 & \text{if } i \text{ survives} \\ Y_i^1 & \text{otherwise} \end{cases} = a_i I_i + Y_i^1$$

$$\Rightarrow \sum_{i=1}^n W_i^1 = \sum_{i=1}^n a_i I_i + \sum_{i=1}^n \sum_{k=1}^k \lambda_k Y_i^{(k)}$$

$$= \sum_{i=1}^n a_i I_i + S_n$$

$$= \sum_{i=1}^n a_i I_i + \sum_{i=1}^n a_i (1 - I_i)$$

$$= \sum_{i=1}^n a_i \quad (\text{Fully-funded})$$

$$\Rightarrow E[W_i^1] = E[a_i I_i] + E[Y_i^1] \quad \leftarrow = E[X_i] = E[a_i] - E[a_i I_i]$$

$$= E[a_i] \leftarrow \text{Expected financial-return}$$

!! Expected gain from pooling is 0.

→ Examples:

I) Conditional mean R-S:

$$Y_i^{(1)} = E[X_i | S_n]$$

II) proportional-mean:

$$Y_i^{(2)} = \frac{E[X_i]}{\sum_{i=1}^n E[X_i]} S_n$$

III) Linear regression rule:

participant may agree to share the total risk according to the relative volatility (especially in the case of stochastic investments).

Let $\sigma_i = \sqrt{\text{Var}(X_i)}$

$$Y_i^{(3)} = E[X_i] + \frac{\sigma_i^2}{\sum_{j=1}^n \sigma_j^2} (S_n - E[S_n])$$

IV) Full Risk-retention:

$$Y_i^{(4)} = X_i \quad (\text{no pooling}).$$

V) Pure endowment sharing-rule:

Inspired by the discussion in "Egalitarian pooling and sharing of longevity risk" article.

$$Y_i^{(5)} = E[X_i] + \frac{c_i/p_i}{\sum_{j=1}^n c_j/p_j} (S_n - E[S_n])$$

where p_i is the survival proba of agent i .

Ⓢ This rule doesn't take into account the investment risk of each participant.

→ Start with individual optimization.

$$\text{Let } Y_i^s = s Y_i^{(1)} + (1-s) Y_i^{(2)} \quad (\text{or } s \in [0,1])$$

$$\max_{s \in [0,1]} E \left[U_i(a_i + Y_i^s) I_i + \beta U_i(Y_i^s) (1 - I_i) \right]$$